

Taco Sales Forecasting & Insights Dashboard

Dataset (Time Series + ML)

Taco Sales Dataset (2024–2025) – Available on Kaggle, this dataset includes recent sales data for tacos throughout 2024 and into 2025. It's ideal for exploratory data analysis, time series forecasting, price prediction, and delivery optimization modeling. [Kaggle](#)

Problem Statement

Objective: Build and deliver a forecasting and analytics solution for taco sales to help a fictional restaurant chain optimize inventory planning, pricing, and promotion strategies.

1. Data Ingestion & Cleaning

- Load and explore the taco sales dataset using Python (e.g., pandas).
- Handle missing values, seasonal patterns, price variations, and any anomalies.

2. Exploratory Data Analysis (EDA)

- Visualize trends in sales volume, revenue, pricing, and day-of-week or seasonal effects using Python (matplotlib, seaborn).

3. Time Series Forecasting

- Develop and compare forecasting models:
 - Traditional: ARIMA, SARIMA, Exponential Smoothing.
 - ML-based: Random Forests or Gradient Boosting on lag features.
 - Advanced: LSTM neural networks or Prophet (if suitable).
- Evaluate model performance with MAE, RMSE, and forecast accuracy metrics.

4. Price Optimization Prediction

- Train a supervised machine learning model (e.g., Random Forest, XGBoost) to predict optimal pricing to maximize sales or revenue, based on historical sales and external factors (e.g., day of week, promotions).

5. Interactive Business Dashboard

- Connect Python insights to **Power BI** or **Tableau** to develop an interactive dashboard featuring:
 - Forecasted sales for the upcoming weeks/months.
 - Historical sales and pricing trends.
 - A tool for adjusting pricing/promotions and viewing forecasted impact in real time.
 - KPI visualizations: projected revenue, stock needs, price elasticity insights.

Why This Project Works

- **Latest data** (2024–2025) ensures relevance and recency. [Kaggle](#)
- **Integrates all requested tools:** Python for data manipulation and modeling, ML for forecasting and prediction, and Power BI/Tableau for interactive visualization and storytelling.
- **Business relevance:** Forecasting and price optimization are highly valuable skills applicable across retail, F&B, and operations.
- **End-to-end scope:** Covers data ingestion, modeling, evaluation, and presentation—all components expected in a capstone.

Project Outline Table

Phase	Tools & Techniques
Data Loading & Cleaning	Python (pandas, numpy)
EDA & Visualization	Python visual libraries (matplotlib, seaborn)
Forecasting	ARIMA/SARIMA, Prophet, LSTM, Random Forests
Price Prediction	ML models: Random Forest, XGBoost
Dashboard Development	Power BI or Tableau, connected via CSV or Python data export

Deliveries

1. Keep all the files in a folder (.zip format)
2. Notebook
3. Ppt
4. Cleaned data if saved
5. Dashboard link attached in the ppt.